Práctica 2

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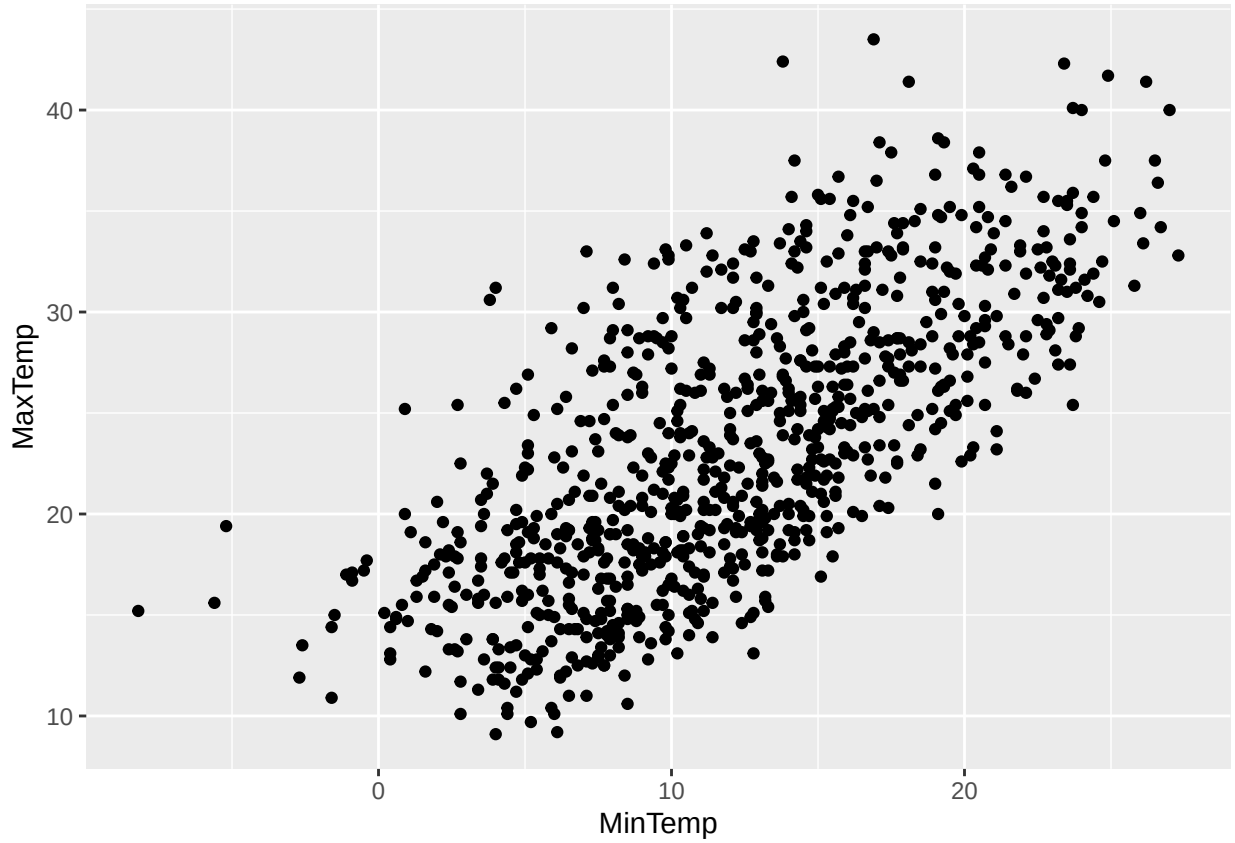
Ejercicio 1

```
load('data/weatherAUS.RData')
str(ds)
```

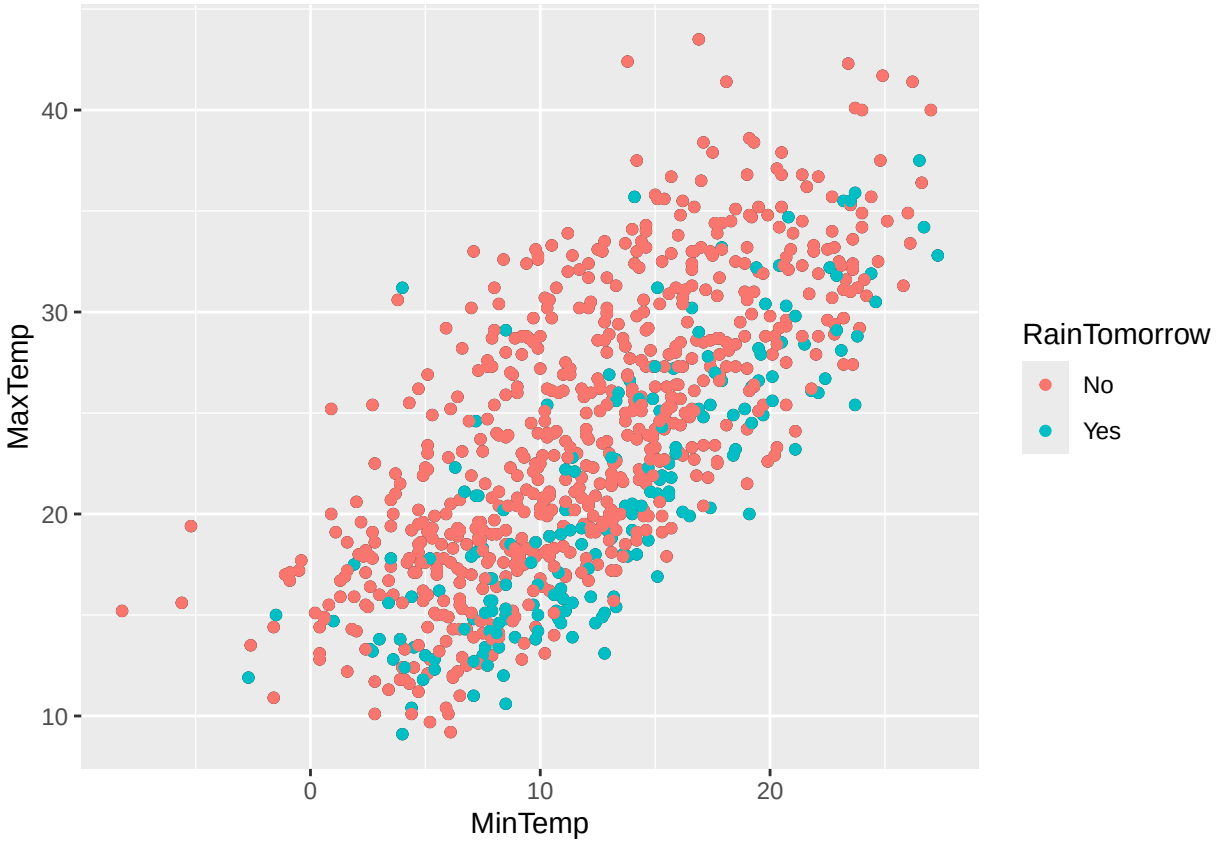
```
## 'data.frame':    863 obs. of  24 variables:
##  $ Date         : Date, format: "2009-03-23" "2014-12-08" ...
##  $ Location     : chr  "Richmond" "Albury" "Launceston" "Sale" ...
##  $ MinTemp      : num  14.2 13.9 0.4 0.2 10.2 9 8.6 24 12.4 24.4 ...
##  $ MaxTemp      : num  29.8 27.7 13.1 15.1 20.7 17.2 23.9 34.2 18 31.9 ...
##  $ Rainfall     : num  0.2 1.4 0 0.2 0 1 0 0 2 18.4 ...
##  $ Evaporation  : chr  "12.2" NA NA "1.2" ...
##  $ Sunshine     : chr  NA NA NA "7.3" ...
##  $ WindGustDir  : chr  "NE" "W" "S" "NW" ...
##  $ WindGustSpeed: int  28 37 19 17 NA 44 63 41 NA 31 ...
##  $ WindDir9am   : chr  NA "ENE" NA "WNW" ...
##  $ WindDir3pm   : chr  "E" "NW" "SSE" "NNW" ...
##  $ WindSpeed9am : int  0 6 0 9 6 22 13 15 6 11 ...
##  $ WindSpeed3pm : int  11 22 4 7 13 30 39 15 35 17 ...
##  $ Humidity9am  : int  92 74 98 93 64 71 62 71 84 83 ...
##  $ Humidity3pm  : int  36 42 57 57 40 63 22 56 61 68 ...
##  $ Pressure9am  : num  1017 1008 1028 1024 1025 ...
##  $ Pressure3pm  : num  1014 1007 1026 1021 1020 ...
##  $ Cloud9am     : int  NA NA NA 2 5 3 2 6 7 6 ...
##  $ Cloud3pm     : int  NA 5 NA 3 7 8 1 3 3 6 ...
##  $ Temp9am      : num  18.3 20.9 4.3 3.7 13 12.8 15.4 29.7 15.3 27.9 ...
##  $ Temp3pm      : num  29.5 26.4 11.9 14.3 18.7 14.6 23.5 32.1 17.5 30.9 ...
##  $ RainToday    : chr  "No" "Yes" "No" "No" ...
##  $ RISK_MM      : num  0 0 0.4 0 0 0.2 0 0 1.6 24.2 ...
##  $ RainTomorrow : chr  "No" "No" "No" "No" ...
##  - attr(*, "spec")=List of 2
##   ..$ cols   :List of 24
##   .. ..$ Date         :List of 1
##   .. .. ..$ format: chr ""
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_date" "collector"
##   .. ..$ Location     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ MinTemp      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ MaxTemp      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Rainfall     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
```

```
##   .. ..$ Evaporation  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ Sunshine     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindGustDir  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindGustSpeed: list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ WindDir9am   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindDir3pm   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindSpeed9am : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ WindSpeed3pm : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Humidity9am  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Humidity3pm  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Pressure9am  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Pressure3pm  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Cloud9am     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Cloud3pm     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Temp9am      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Temp3pm      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ RainToday    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ RISK_MM      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ RainTomorrow : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   ..$ default: list()
##   .. ..- attr(*, "class")= chr [1:2] "collector_guess" "collector"
##   ..- attr(*, "class")= chr "col_spec"
```

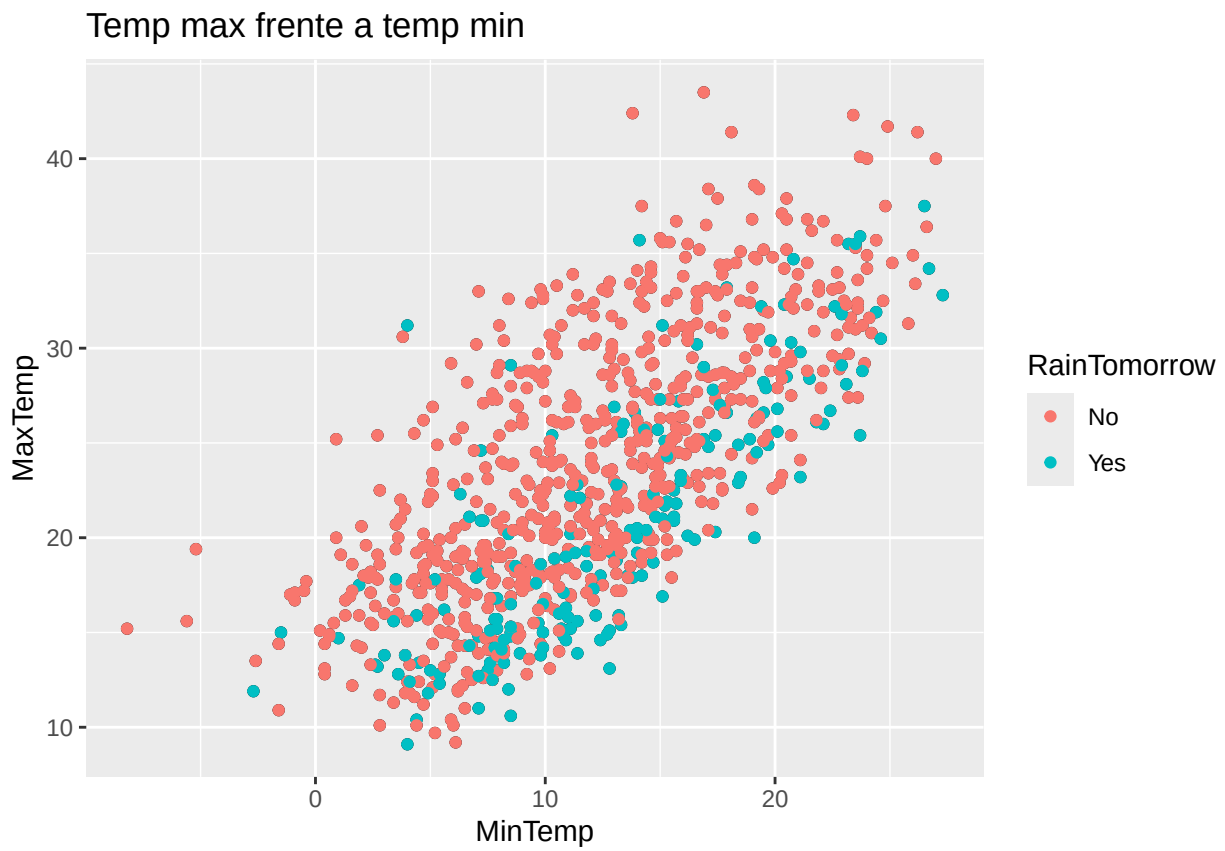
```r
library(ggplot2)
grafico1 <- ggplot(data = ds, mapping = aes(x = MinTemp, y = MaxTemp))
grafico1 <- grafico1 + geom_point()
grafico1
```




```
grafico2 <- grafico1 + geom_point(aes(color = RainTomorrow))
grafico2
```




```
grafico3 <- grafico2 + ggtitle('Temp max frente a temp min')
grafico3
```

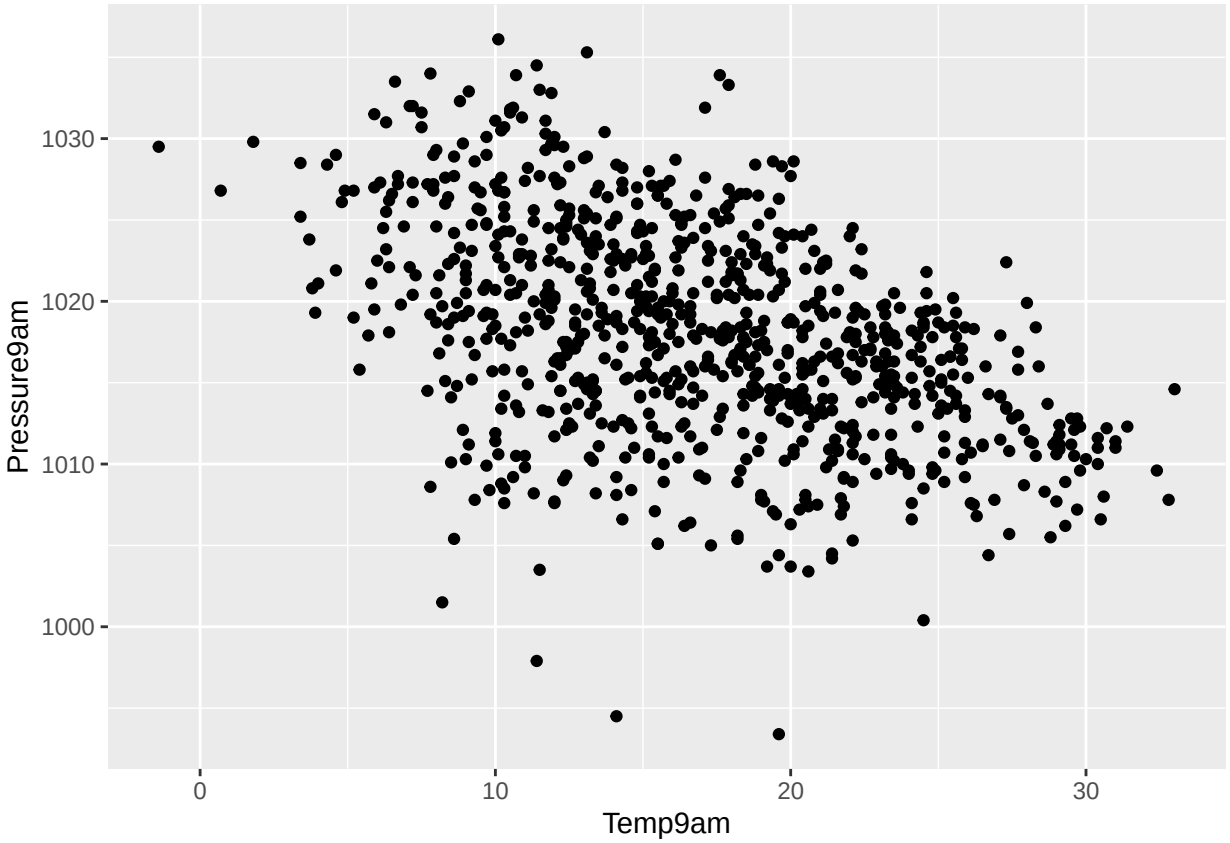



```
if (!file.exists('results')){
  dir.create('results')
}
ggsave('./results/Ejercicio1.pdf')
```

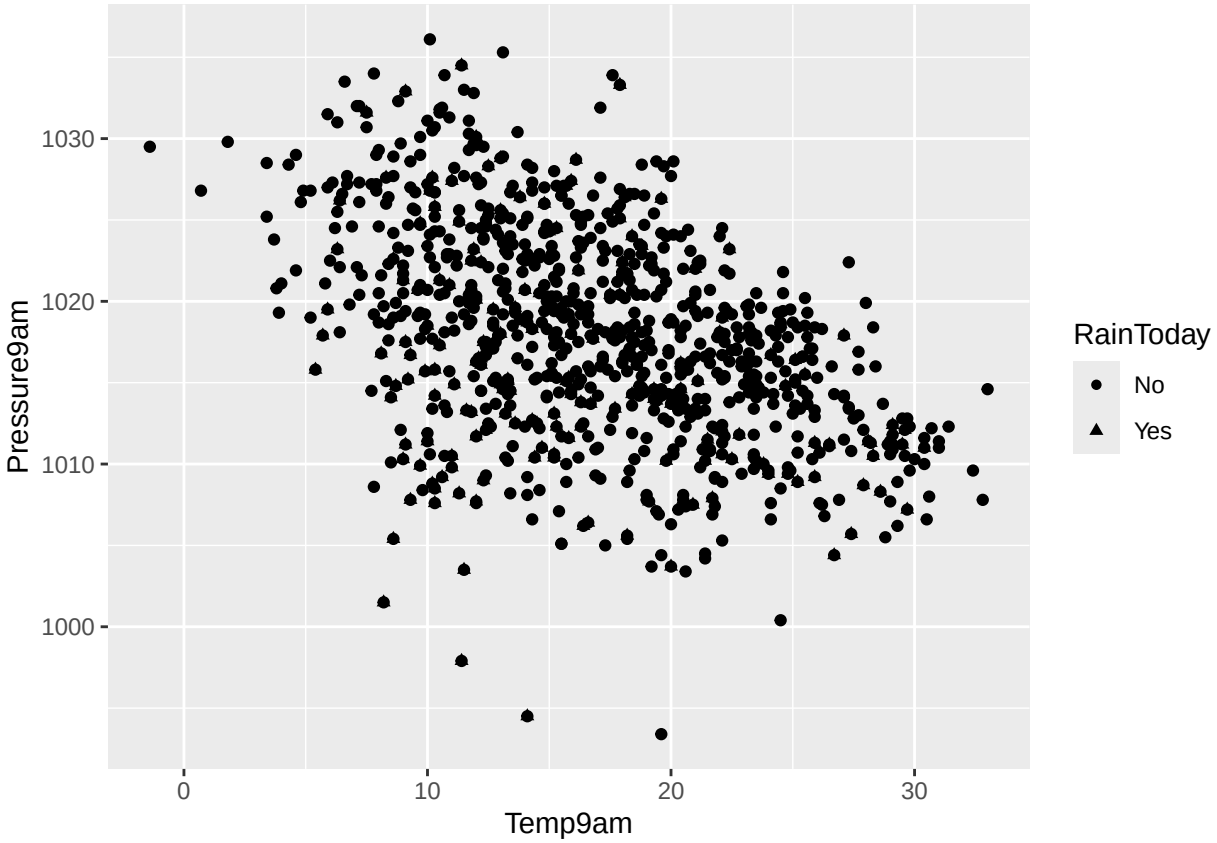
```
## Saving 6.5 x 4.5 in image
```

Ejercicio 2

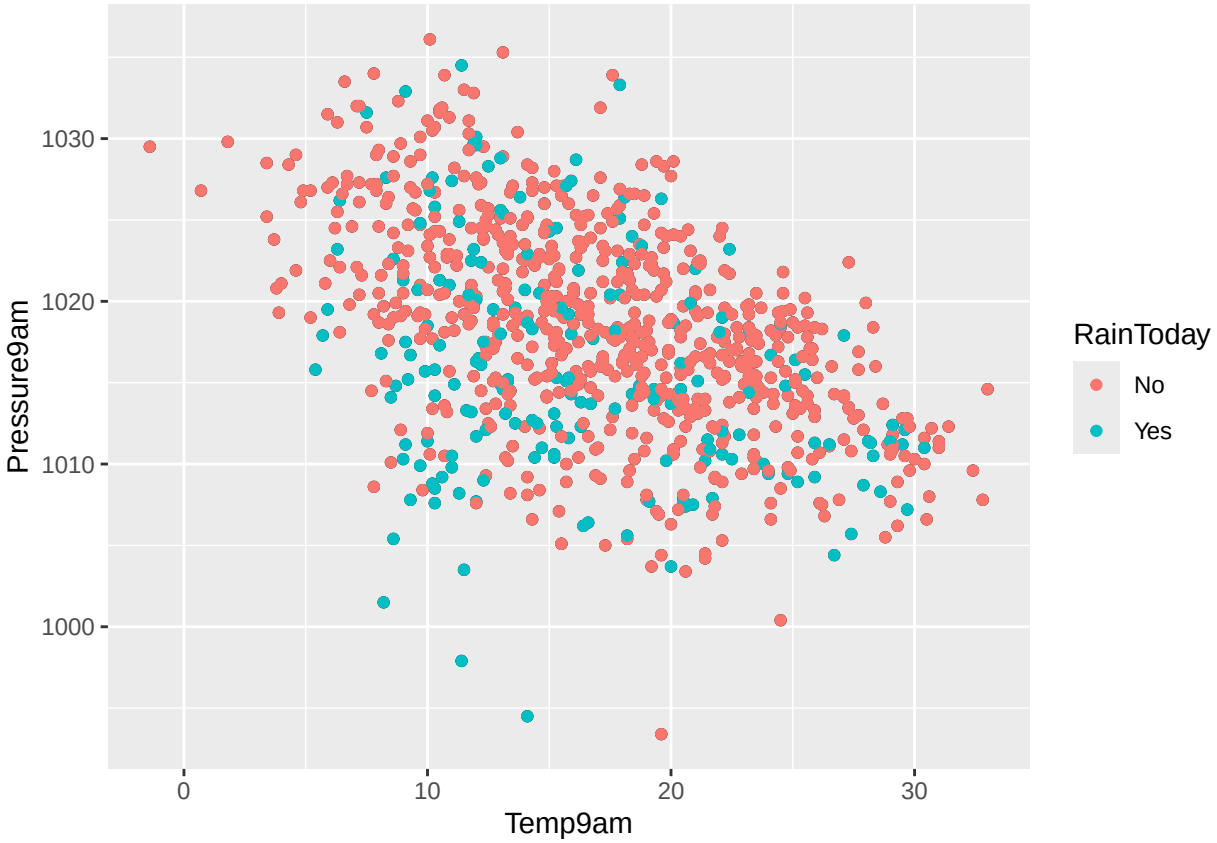
```
grafico9am <- ggplot(data = ds, mapping = aes(x = Temp9am, y = Pressure9am)) + geom_point()
grafico9am
```




```
grafico9am.punto <- grafico9am + geom_point(aes(pch = RainToday))
grafico9am.punto
```

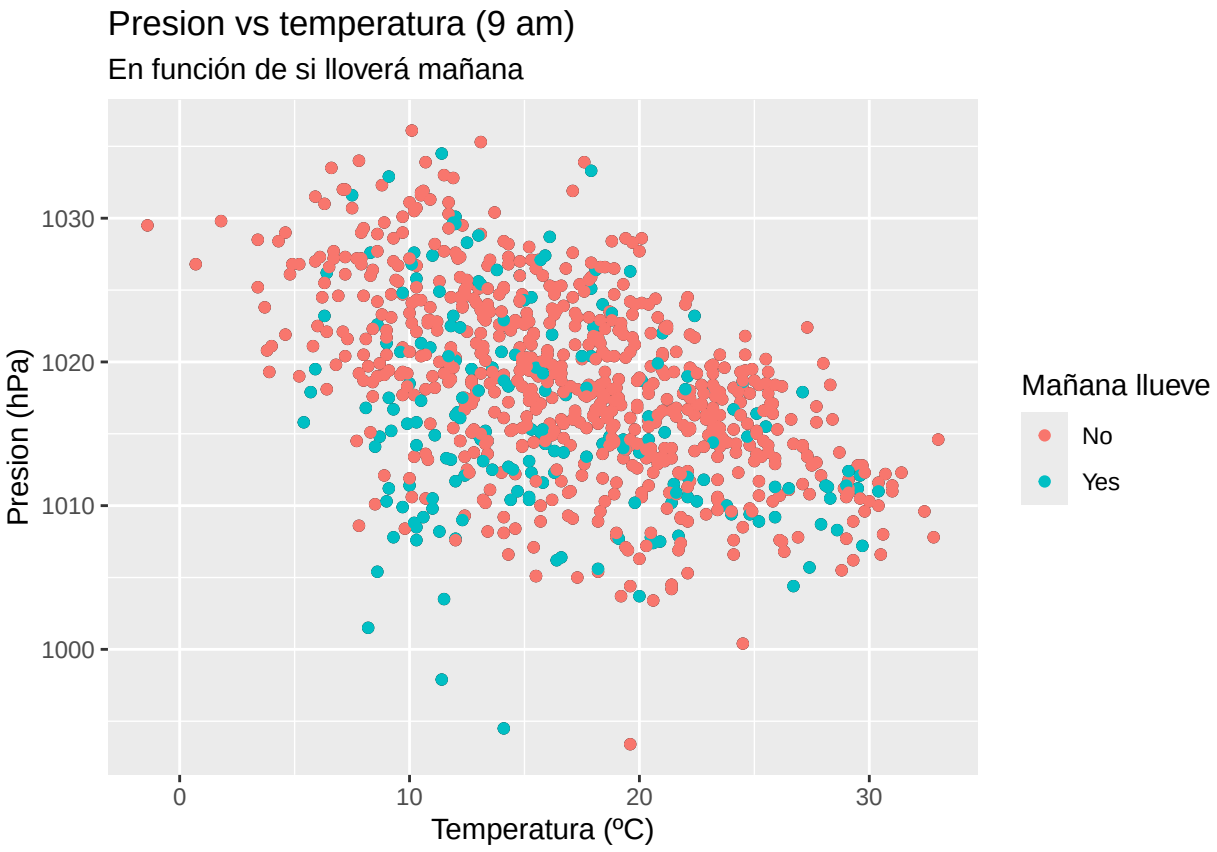



```
grafico9am.color <- grafico9am + geom_point(aes(color = RainToday))
grafico9am.color
```



El color permite ver mejor la clasificación de los puntos, ya que cuando estos están muy cerca la diferencia entre el tipo de punto (pch) no es fácilmente distinguible.

```
grafico9am.color <- grafico9am.color + labs(title = 'Presion vs temperatura (9 am)', subtitle = 'En fun
grafico9am.color
```




```
ggsave('./results/Ejercicio2.pdf')
```

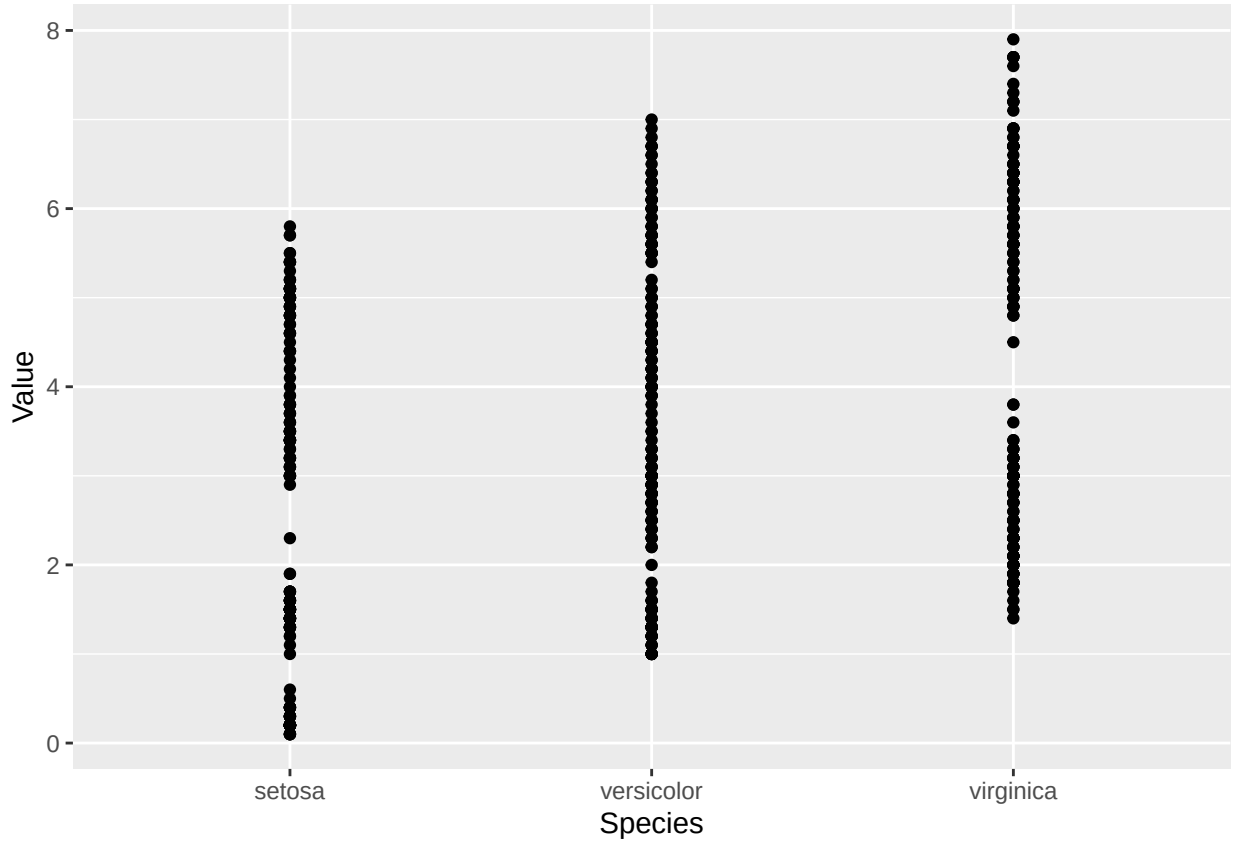
```
## Saving 6.5 x 4.5 in image
```

Ejercicio 3

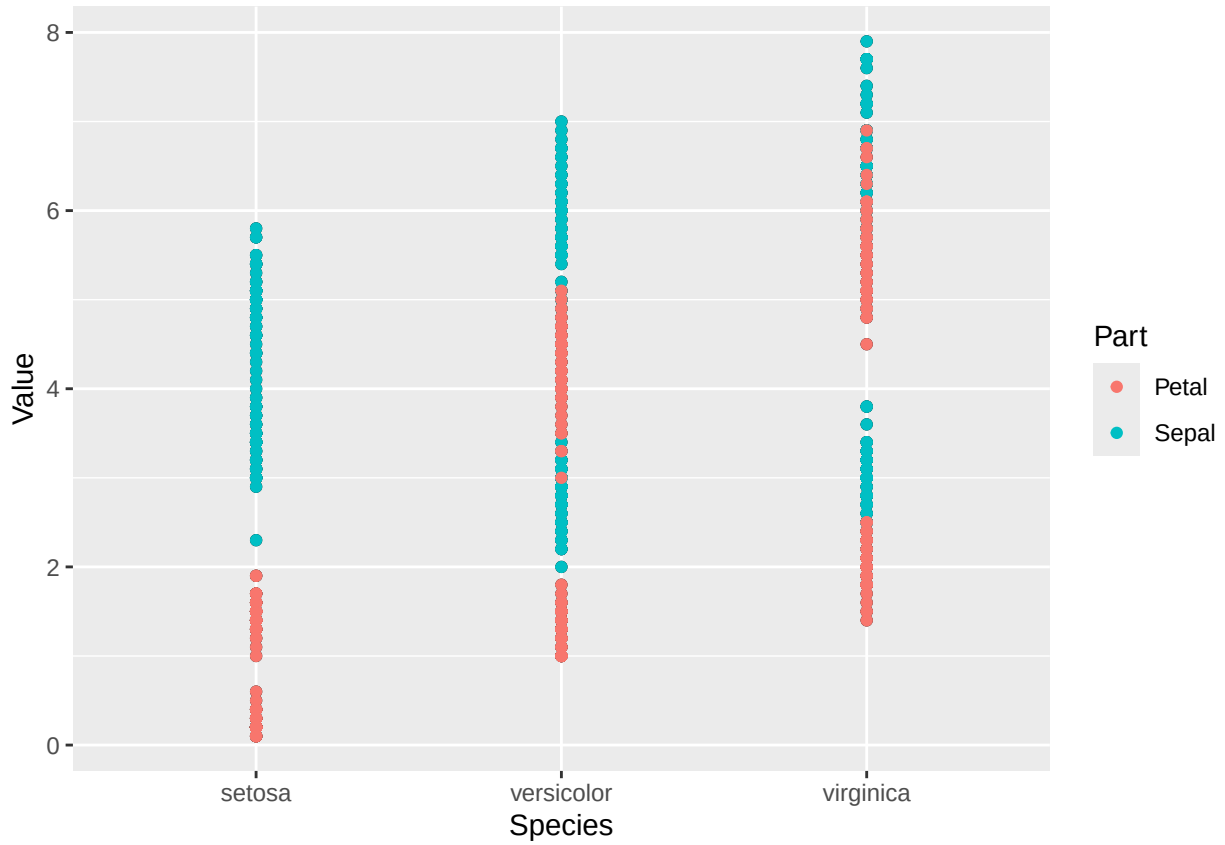
```
load('./data/iris_tidy.RData')
str(iris.tidy)
```

```
## 'data.frame':    600 obs. of  4 variables:
##  $ Species: Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ Part   : Factor w/ 2 levels "Petal","Sepal": 2 2 2 2 2 2 2 2 2 2 ...
##  $ Measure: chr  "Length" "Length" "Length" "Length" ...
##  $ Value  : num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
```

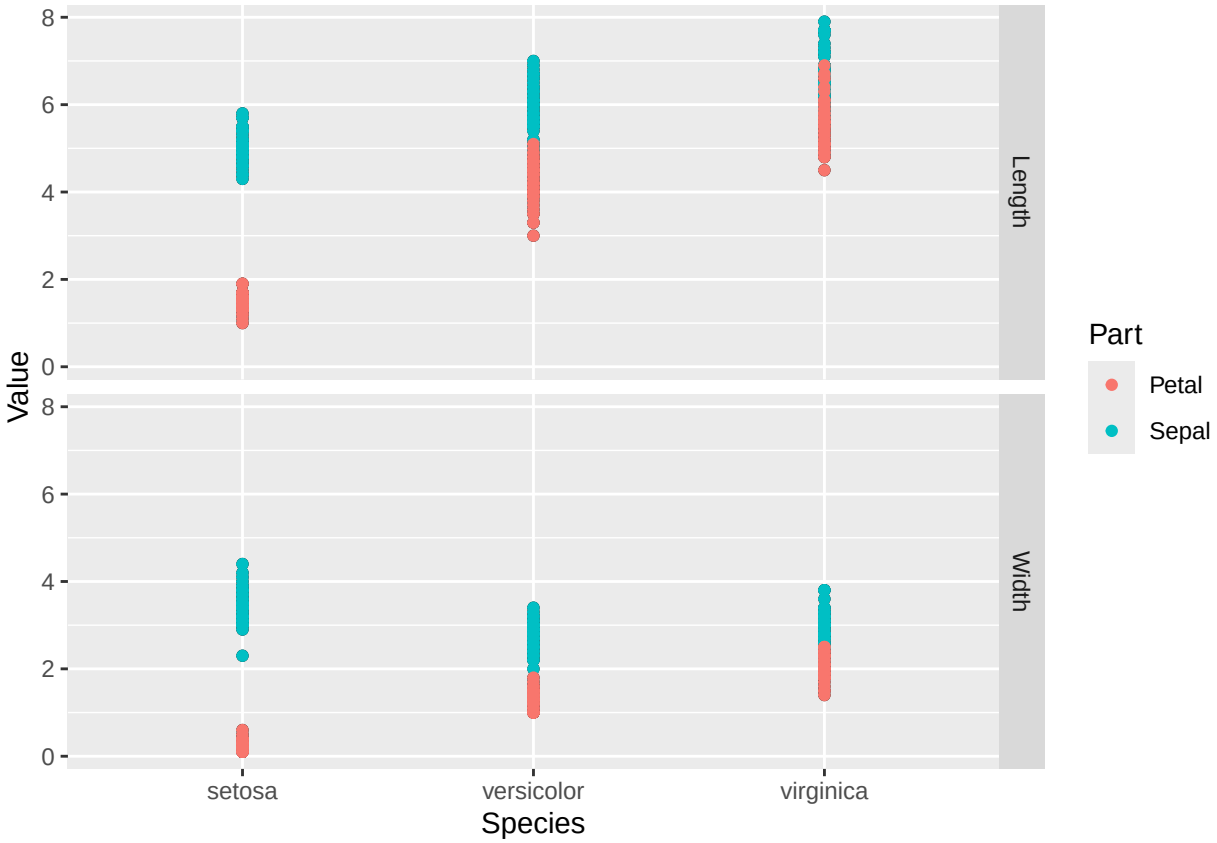
```
iris.grafico1 <- ggplot(data = iris.tidy, mapping = aes(x = Species, y = Value)) + geom_point()
iris.grafico1
```



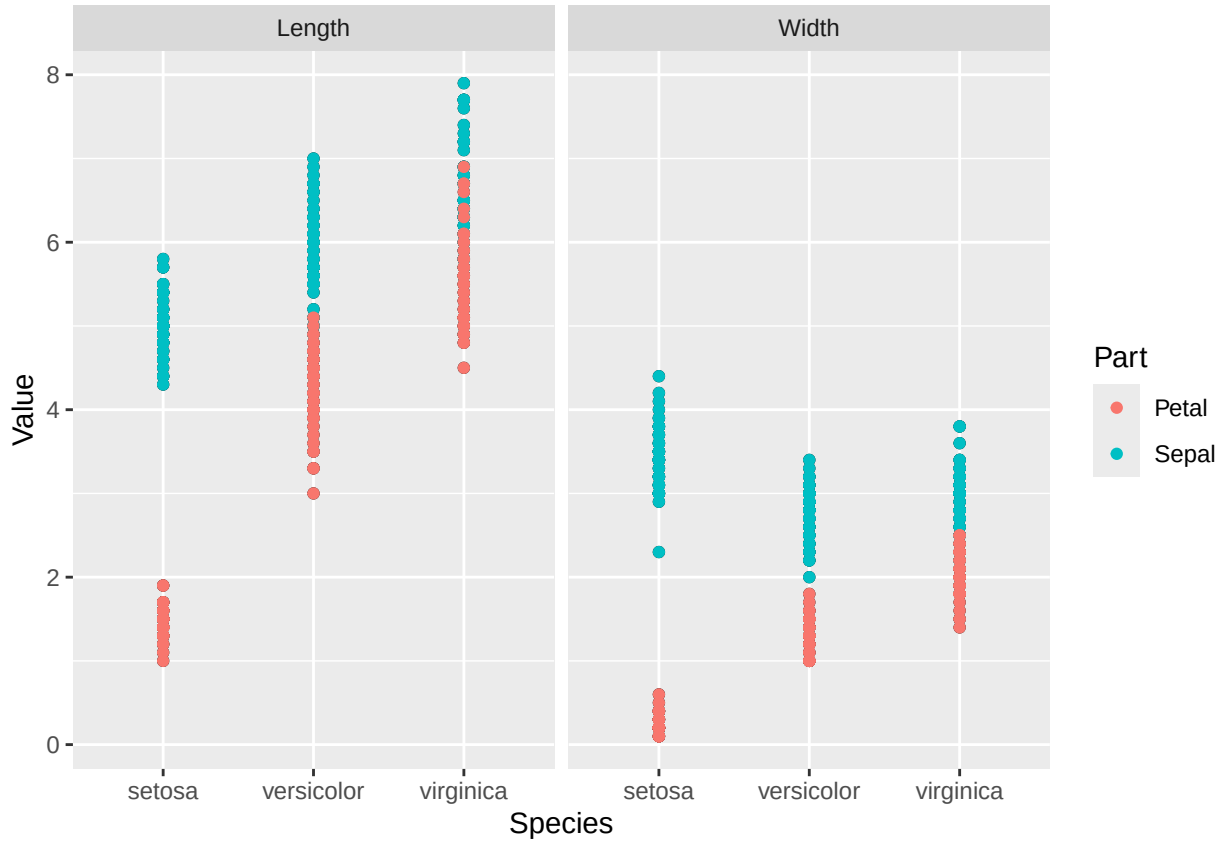

```
iris.grafico2 <- iris.grafico1 + geom_point(mapping = aes(color = Part))
iris.grafico2
```




```
iris.grafico2.filas <- iris.grafico2 + facet_grid(Measure~.)
iris.grafico2.filas
```



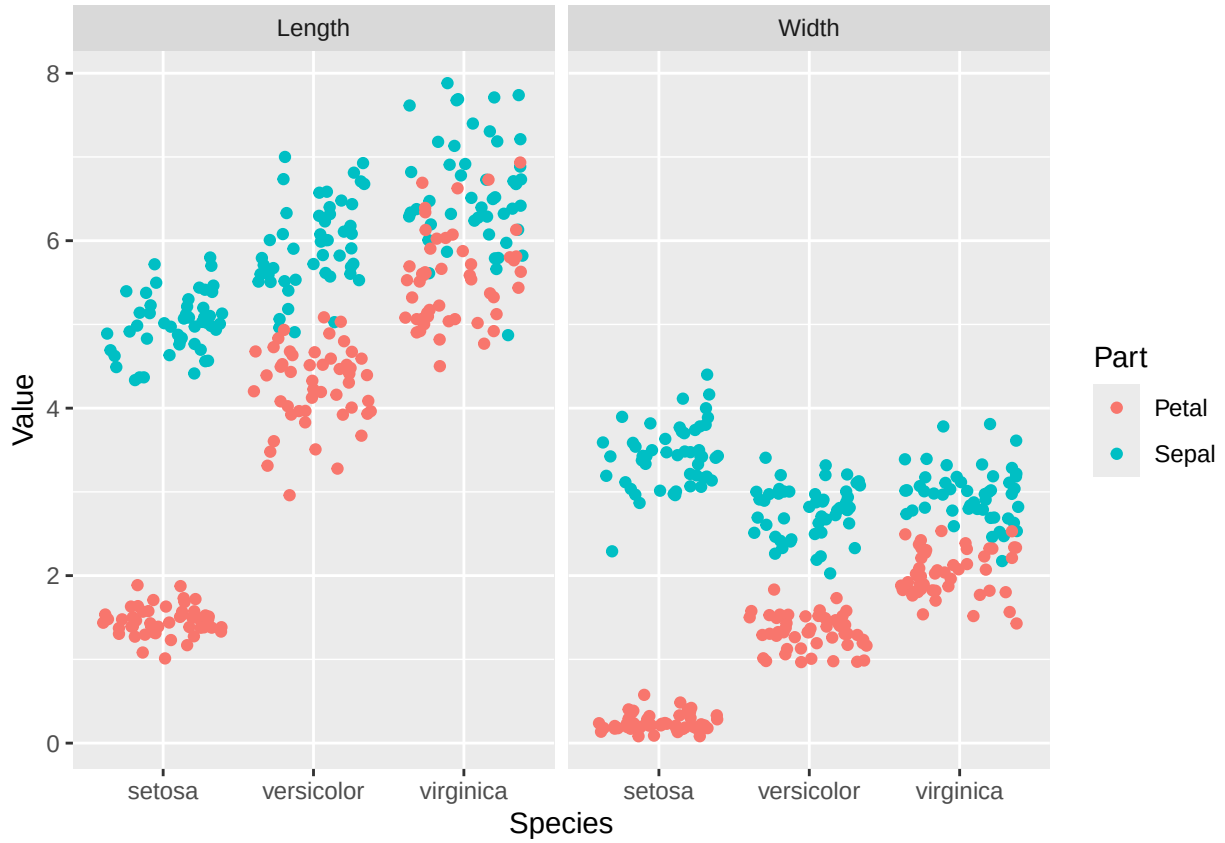

```
iris.grafico2.col <- iris.grafico2 + facet_grid(.~Measure)
iris.grafico2.col
```



La distribución en columnas queda más clara, ya que la distribución de puntos es en bandas verticales, debido a la naturaleza categórica de la variable Species.

```
# En algun momento del codigo hemos definido 2 capas de geom_point
iris.grafico2.col$layers[[1]] <- NULL # Aqui elimino la primera
iris.grafico2.col$layers[[1]] <- geom_jitter(aes(color = Part))
# Y aqui hago que la 2a pase de point a jitter

iris.grafico2.col
```



Al añadir ruido al gráfico, ahora los puntos dejan de estar superpuestos y se puede ver la cantidad que hay para cada variable.

```
ggsave('./results/Ejercicio3.pdf')
```

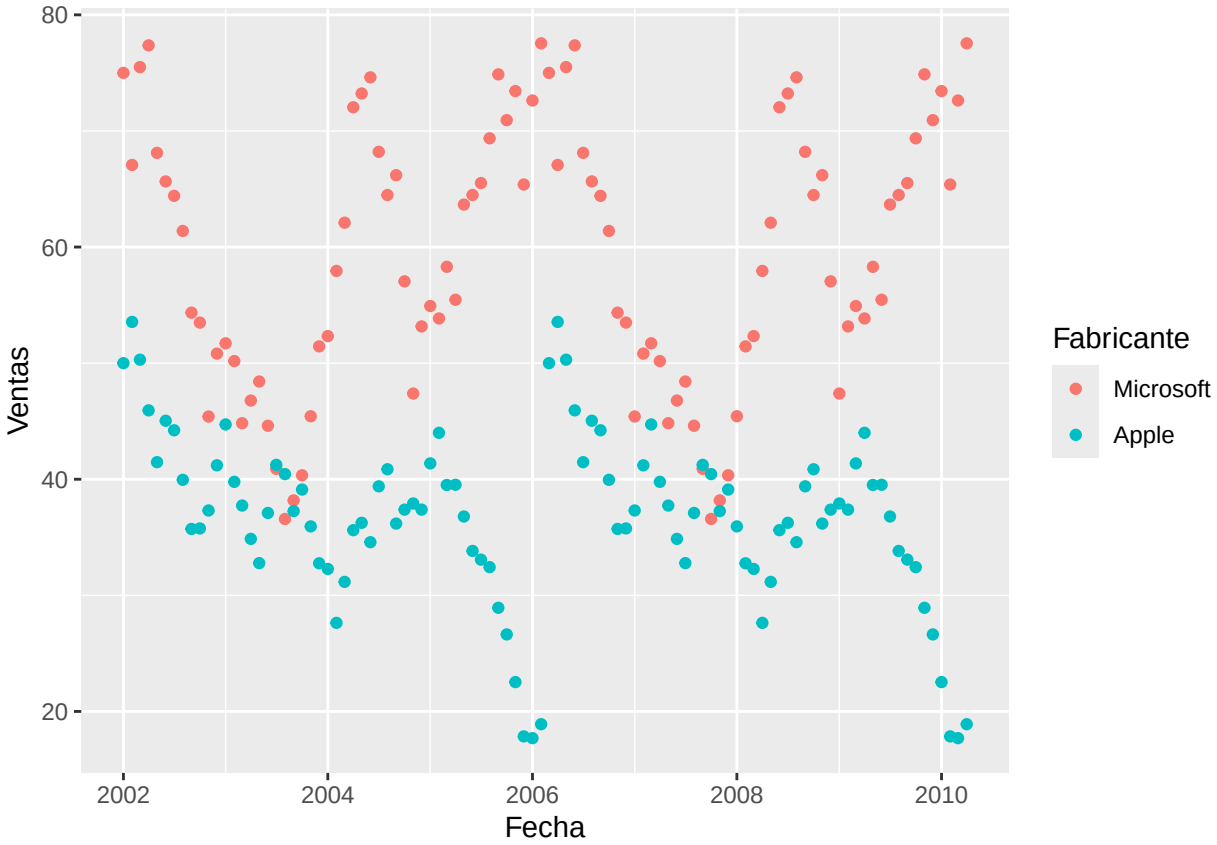
```
## Saving 6.5 x 4.5 in image
```

Ejercicio 4

```
load('./data/Ejercicio4_Data.RData')
str(DatosEjer4)
```

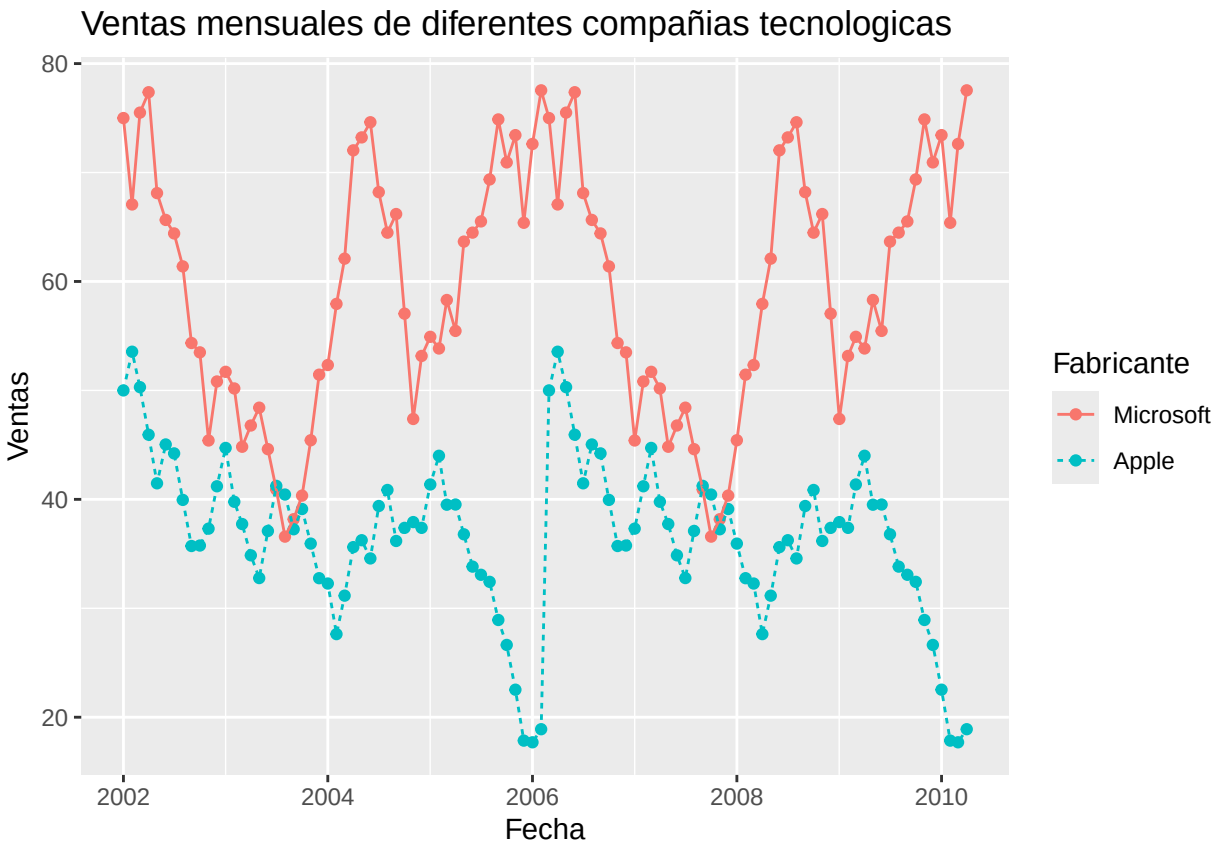
```
## 'data.frame':    200 obs. of  3 variables:
##  $ Fecha     : Date, format: "2002-01-01" "2002-02-01" ...
##  $ Fabricante: Factor w/ 2 levels "Microsoft","Apple": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Ventas    : num  75 67.1 75.5 77.4 68.1 ...
```

```
graf4 <- ggplot(data = DatosEjer4, mapping = aes(x = Fecha, y = Ventas)) + geom_point(aes(color = Fabri
graf4
```



Es preferible un gráfico de líneas, ya que solo usar puntos añade ruido y confusión al gráfico, al no poderse leer bien la evolución temporal.

```
graf4 <- graf4 + geom_line(aes(linetype = Fabricante, color = Fabricante)) + labs(title  ='Ventas mensua
graf4
```




```
ggsave('./results/Ejercicio4.pdf')
```

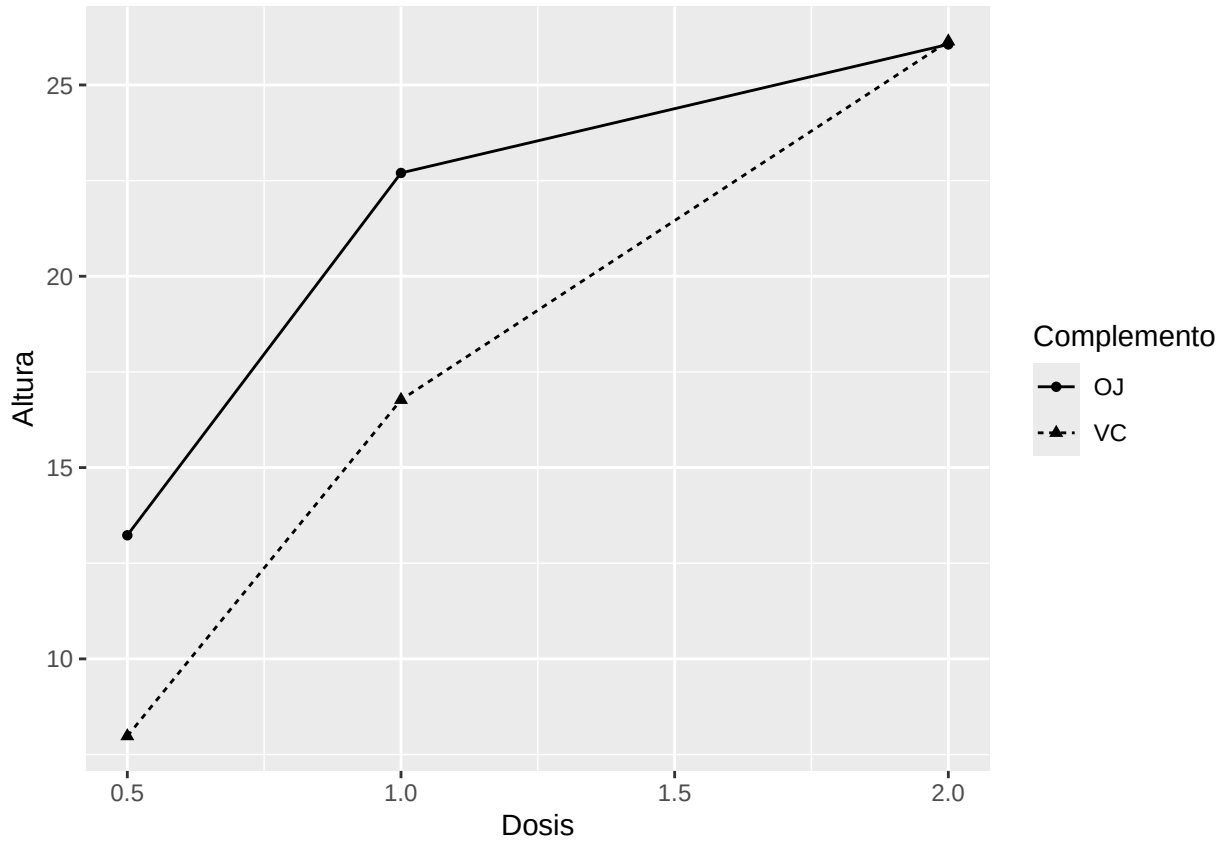
```
## Saving 6.5 x 4.5 in image
```

Ejercicio 5

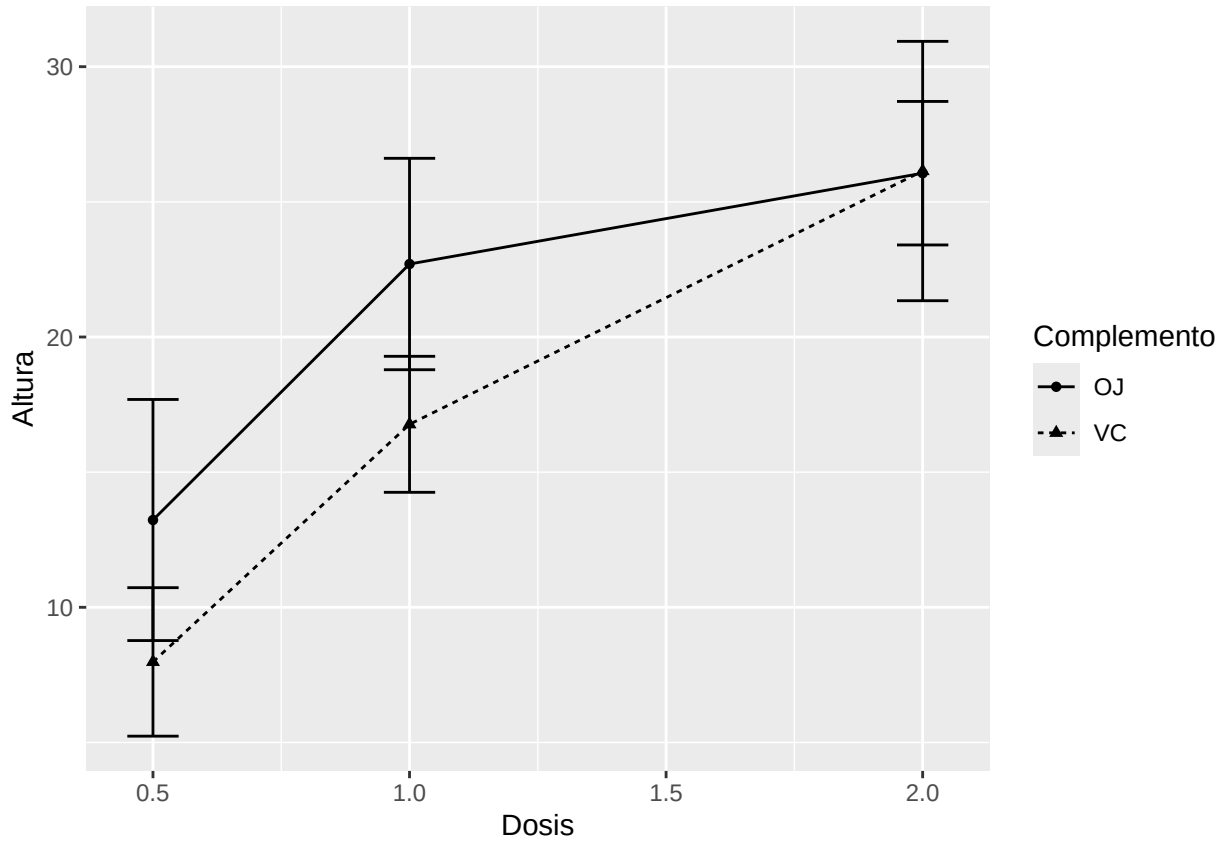
```
load('./data/Experimento.RData')
str(Experimento)
```

```
## 'data.frame':    6 obs. of  4 variables:
##  $ Complemento : Factor w/ 2 levels "OJ","VC": 1 1 1 2 2 2
##  $ Dosis       : num  0.5 1 2 0.5 1 2
##  $ Altura      : num  13.23 22.7 26.06 7.98 16.77 ...
##  $ Error_Altura: num  4.46 3.91 2.66 2.75 2.52 ...
```

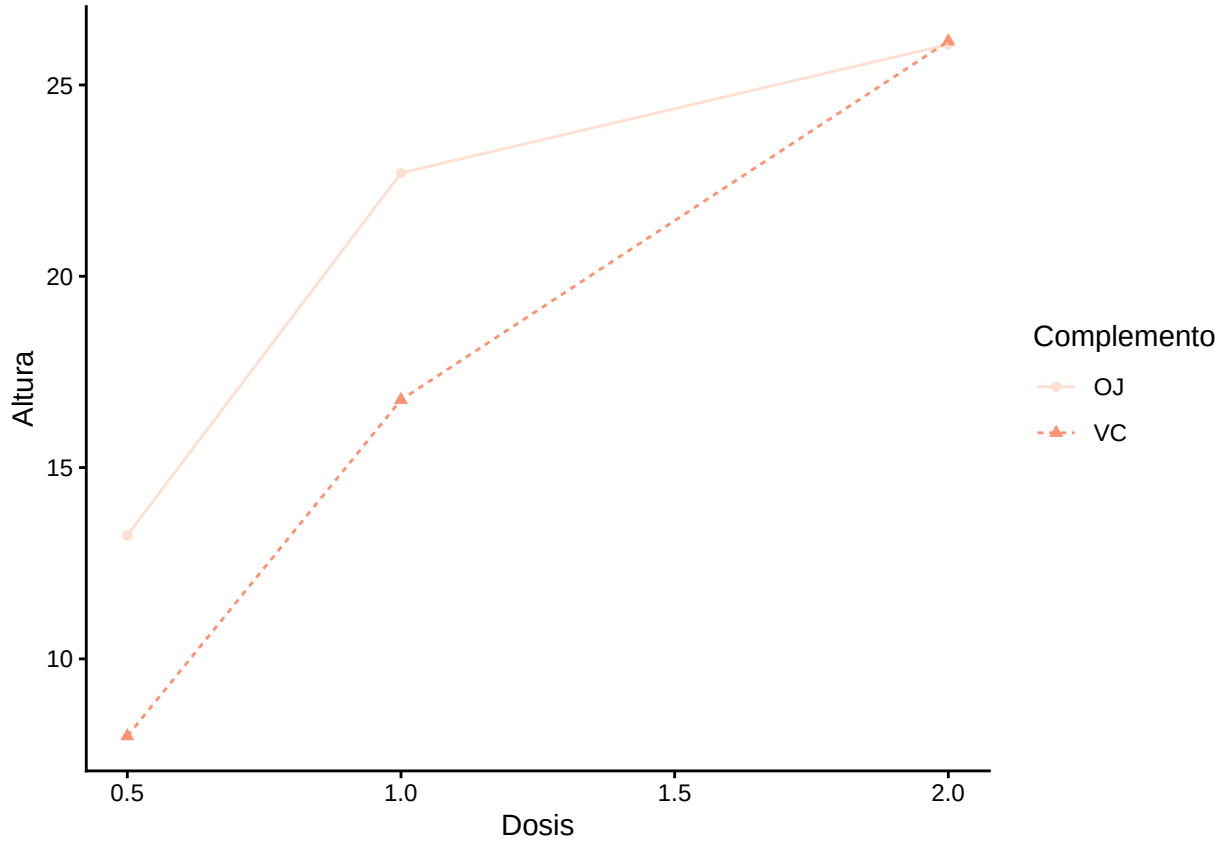
```
graf.exp <- ggplot(data = Experimento, mapping = aes(x = Dosis, y = Altura)) + geom_point(aes(pch = Comp
graf.exp
```




```
graf.exp <- graf.exp + geom_errorbar(aes(ymin = Altura - Error_Altura, ymax = Altura + Error_Altura), wi
graf.exp
```




```
graf.exp2 <- ggplot(data = Experimento, mapping = aes(x = Dosis, y = Altura, color = Complemento, pch =
# Quitamos el fondo con theme_classic para que se vea mejor
graf.exp2
```




```
ggsave('./results/Ejercicio5.pdf')
```

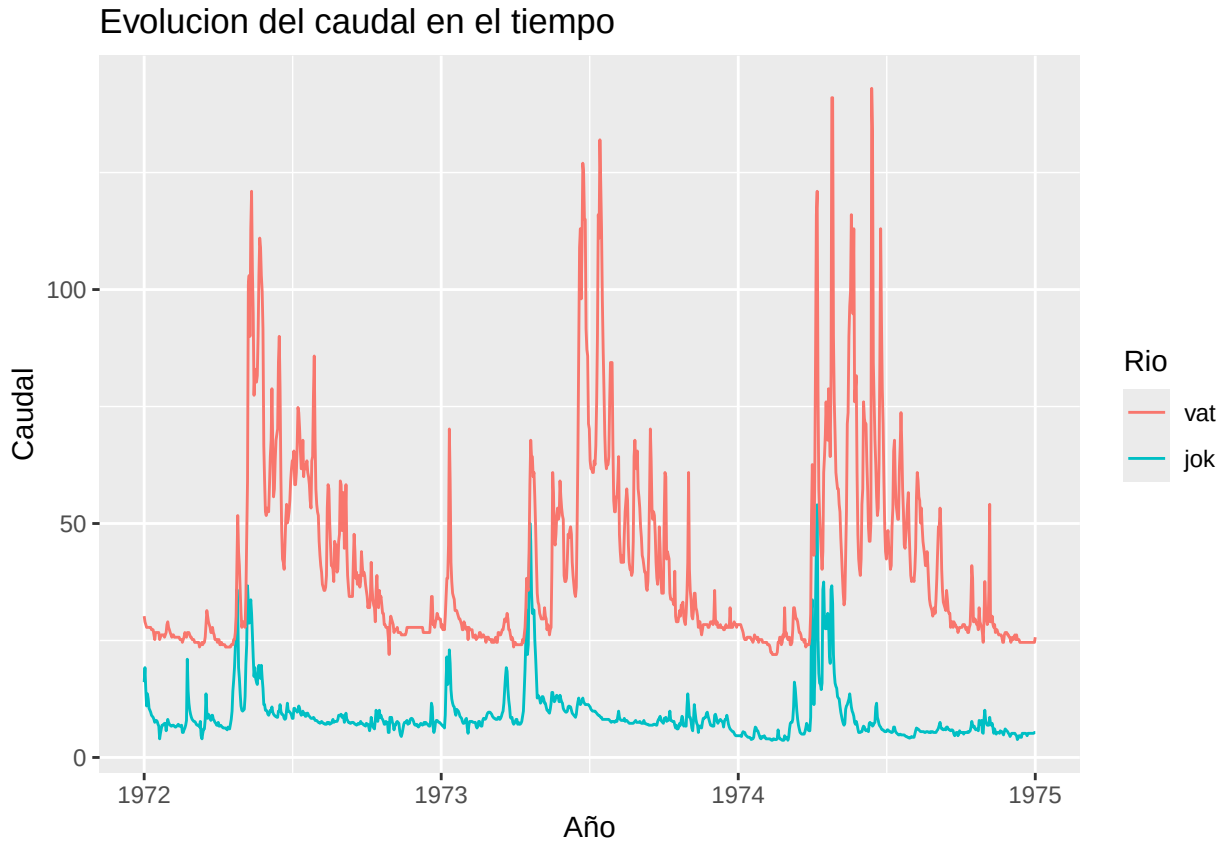
```
## Saving 6.5 x 4.5 in image
```

Ejercicio 6

```
load('./data/iceRiver.RData')
str(ice.river.df)
```

```
## 'data.frame':    1096 obs. of  5 variables:
##  $ flow.vat: num  16.1 19.2 14.5 11 13.6 12.5 10.5 10.1 9.68 9.02 ...
##  $ flow.jok: num  30.2 29 28.4 27.8 27.8 27.8 27.8 27.8 27.8 27.3 ...
##  $ prec    : num  8.1 4.4 7 0 0 0 1.9 1.2 0 0.1 ...
##  $ temp    : num  0.9 1.6 0.1 0.6 2 0.8 1.4 1.3 2.2 0.1 ...
##  $ time    : num  1972 1972 1972 1972 1972 ...
```

```
graf.flow <- ggplot(data = ice.river.df) + geom_line(aes(x = time, y = flow.vat, color = 'red')) + geom_
graf.flow
```




```
ggsave('./results/Ejercicio6.pdf')
```

```
## Saving 6.5 x 4.5 in image
```

Ejercicio 7

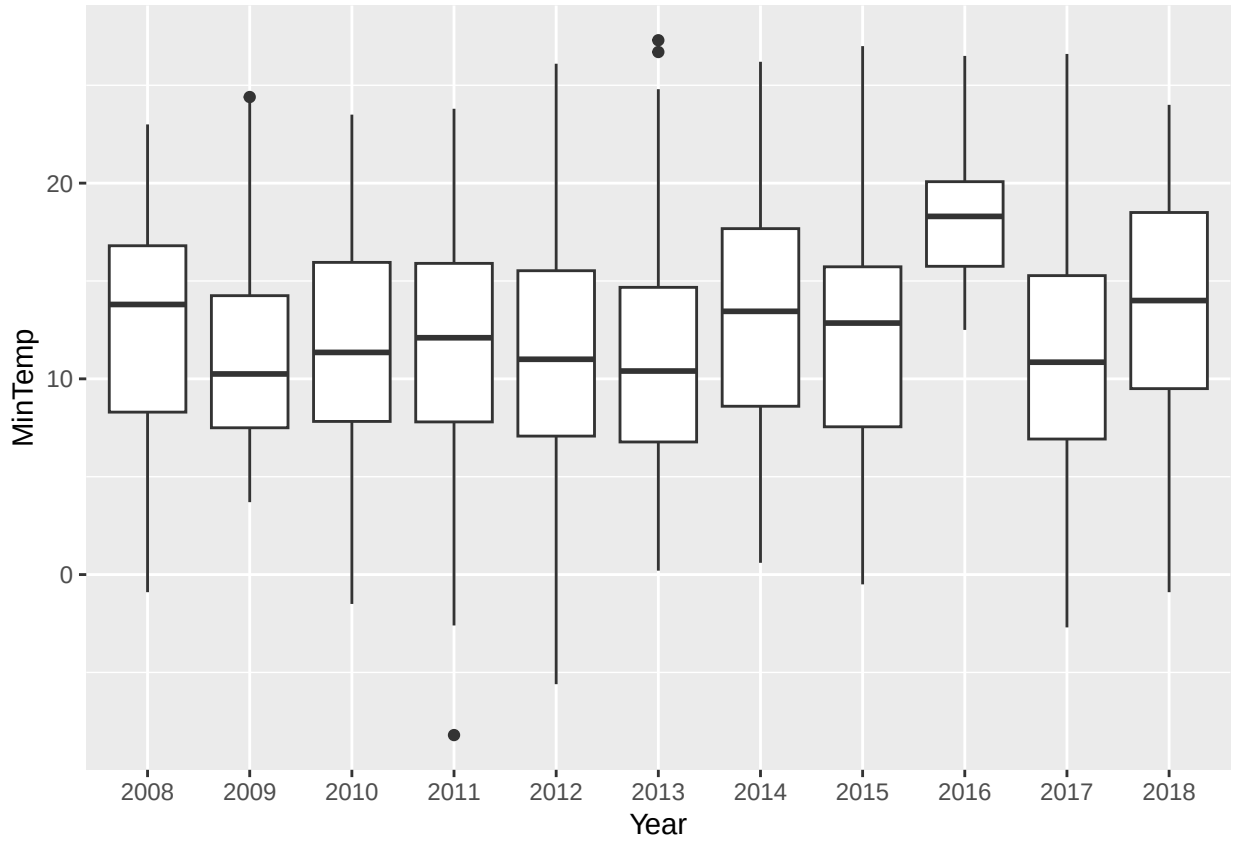
```
load('./data/weatherAUS_distributions.RData')
str(ds)
```

```
## 'data.frame':    863 obs. of  24 variables:
##  $ Year         : Factor w/ 11 levels "2008","2009",..: 2 7 3 6 5 10 10 7 7 2 ...
##  $ Location     : chr  "Richmond" "Albury" "Launceston" "Sale" ...
##  $ MinTemp      : num  14.2 13.9 0.4 0.2 10.2 9 8.6 24 12.4 24.4 ...
##  $ MaxTemp      : num  29.8 27.7 13.1 15.1 20.7 17.2 23.9 34.2 18 31.9 ...
##  $ Rainfall     : num  0.2 1.4 0 0.2 0 1 0 0 2 18.4 ...
##  $ Evaporation  : chr  "12.2" NA NA "1.2" ...
##  $ Sunshine     : chr  NA NA NA "7.3" ...
##  $ WindGustDir  : chr  "NE" "W" "S" "NW" ...
##  $ WindGustSpeed: int  28 37 19 17 NA 44 63 41 NA 31 ...
##  $ WindDir9am   : chr  NA "ENE" NA "WNW" ...
##  $ WindDir3pm   : chr  "E" "NW" "SSE" "NNW" ...
##  $ WindSpeed9am : int  0 6 0 9 6 22 13 15 6 11 ...
##  $ WindSpeed3pm : int  11 22 4 7 13 30 39 15 35 17 ...
```

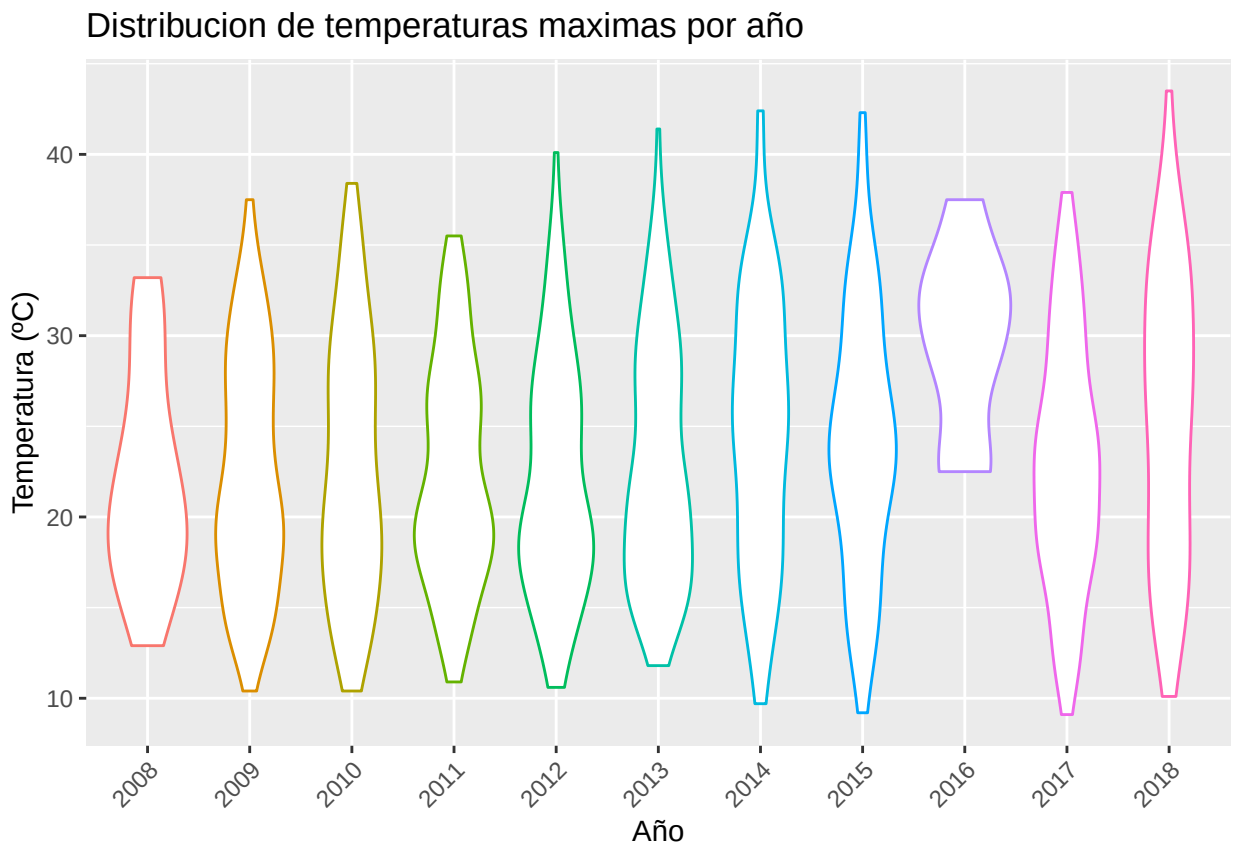
```
##  $ Humidity9am  : int  92 74 98 93 64 71 62 71 84 83 ...
##  $ Humidity3pm  : int  36 42 57 57 40 63 22 56 61 68 ...
##  $ Pressure9am  : num  1017 1008 1028 1024 1025 ...
##  $ Pressure3pm  : num  1014 1007 1026 1021 1020 ...
##  $ Cloud9am     : int  NA NA NA 2 5 3 2 6 7 6 ...
##  $ Cloud3pm     : int  NA 5 NA 3 7 8 1 3 3 6 ...
##  $ Temp9am      : num  18.3 20.9 4.3 3.7 13 12.8 15.4 29.7 15.3 27.9 ...
##  $ Temp3pm      : num  29.5 26.4 11.9 14.3 18.7 14.6 23.5 32.1 17.5 30.9 ...
##  $ RainToday    : chr  "No" "Yes" "No" "No" ...
##  $ RISK_MM      : num  0 0 0.4 0 0 0.2 0 0 1.6 24.2 ...
##  $ RainTomorrow : chr  "No" "No" "No" "No" ...
##  - attr(*, "spec")=List of 2
##   ..$ cols   :List of 24
##   .. ..$ Date         :List of 1
##   .. .. ..$ format: chr ""
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_date" "collector"
##   .. ..$ Location     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ MinTemp      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ MaxTemp      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Rainfall     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Evaporation  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ Sunshine     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindGustDir  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindGustSpeed: list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ WindDir9am   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindDir3pm   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ WindSpeed9am : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ WindSpeed3pm : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Humidity9am  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Humidity3pm  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Pressure9am  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Pressure3pm  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ Cloud9am     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Cloud3pm     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_integer" "collector"
##   .. ..$ Temp9am      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
```

```
##   .. ..$ Temp3pm      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ RainToday    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   .. ..$ RISK_MM      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ RainTomorrow : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_character" "collector"
##   ..$ default: list()
##   .. ..- attr(*, "class")= chr [1:2] "collector_guess" "collector"
##   ..- attr(*, "class")= chr "col_spec"
```

```
graf7 <- ggplot(data = ds) + geom_boxplot(aes(y = MinTemp, x = Year))
graf7
```




```
graf7vi <- ggplot(data = ds) + geom_violin(aes(x = Year, y = MaxTemp, color = Year)) + labs(title = 'Di
graf7vi
```




```
ggsave('./results/Ejercicio7.pdf')
```

```
## Saving 6.5 x 4.5 in image
```

Ejercicio 8

```
library(readr)
ejer8 <- read_delim('http://assets.datacamp.com/blog_assets/chol.txt', delim = ' ')
```

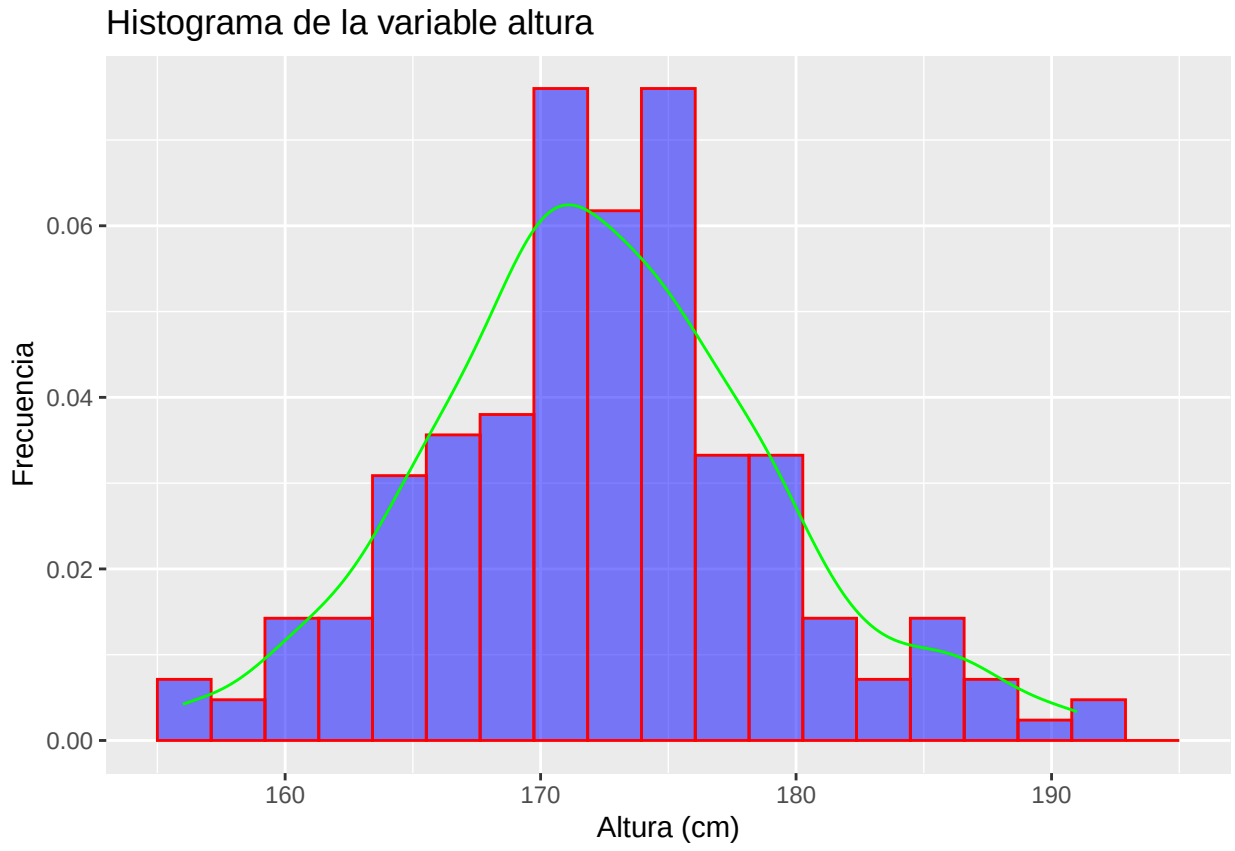
```
## Rows: 200 Columns: 7
## -- Column specification --------------------------------------------------------
## Delimiter: " "
## chr (3): SMOKE, BLOOD, MORT
## dbl (4): AGE, HEIGHT, WEIGHT, CHOL
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
str(ejer8)
```

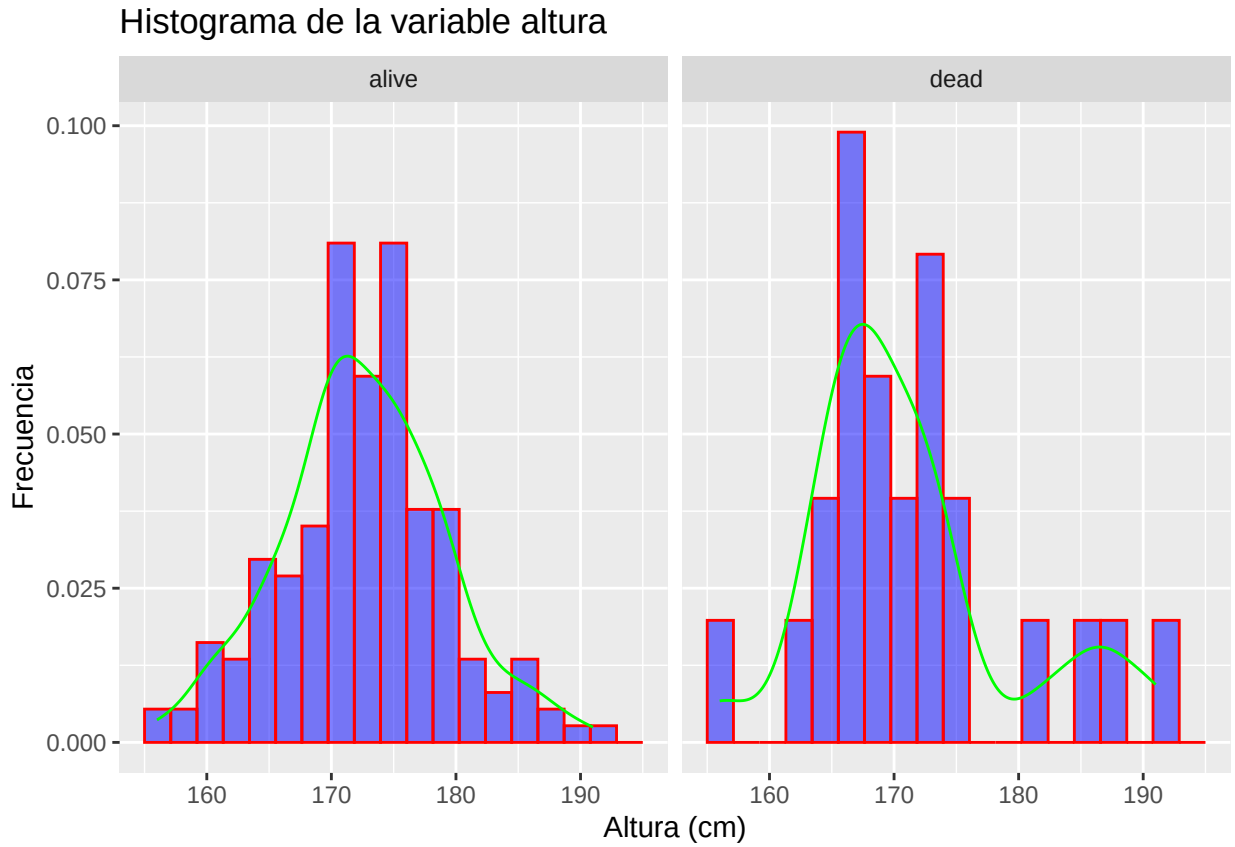
```
## spc_tbl_ [200 x 7] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ AGE    : num [1:200] 20 53 44 37 26 41 39 28 33 39 ...
```

```
##  $ HEIGHT: num [1:200] 176 167 170 173 170 165 174 171 180 166 ...
##  $ WEIGHT: num [1:200] 77 56 80 89 71 62 75 68 100 74 ...
##  $ CHOL  : num [1:200] 195 250 304 178 206 284 232 152 209 150 ...
##  $ SMOKE : chr [1:200] "nonsmo" "sigare" "sigare" "nonsmo" ...
##  $ BLOOD : chr [1:200] "b" "o" "a" "o" ...
##  $ MORT  : chr [1:200] "alive" "dead" "dead" "alive" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   AGE = col_double(),
##   ..   HEIGHT = col_double(),
##   ..   WEIGHT = col_double(),
##   ..   CHOL = col_double(),
##   ..   SMOKE = col_character(),
##   ..   BLOOD = col_character(),
##   ..   MORT = col_character()
##   .. )
##  - attr(*, "problems")=<pointer: 0x000002065aaef680>
```

```
graf8 <- ggplot(data = ejer8, mapping = aes(x = HEIGHT)) + geom_histogram(aes(y = after_stat(density)),
graf8
```




```
graf8mort <- ggplot(data = ejer8, mapping = aes(x = HEIGHT)) + geom_histogram(aes(y = after_stat(densit
graf8mort
```




```
ggsave('./results/Ejercicio8.pdf')
```

```
## Saving 6.5 x 4.5 in image
```